MATHEMATICS 4052 · PAPER 2 · PRACTICE PAPER IN THE GCE FORMAT · WORKED SOLUTIONS

9 questions

1. (a) Solve the inequality $5 - 3(x - 2) > 2x + 1$. [2]

(b) Simplify $\left(9p^4q^{-2}\right)^{\frac{1}{2}} \times 2p^{-3}q^5$, giving your answer in positive index form. [2]

(c) Make t the subject of the formula $v = \sqrt{\frac{k - 3t}{t}}$. [3]

(d) Solve the equation $\frac{4}{x-1} + \frac{3}{x+2} = 2$. Give your solutions correct to two decimal places. [4]

(a) $5 - 3(x - 2) > 2x + 1 \n5 - 3x + 6 > 2x + 1 \n11 - 3x > 2x + 1 \n10 > 5x \nx < 2$.

(b) $\left(9p^4q^{-2}\right)^{\frac{1}{2}} = 9^{\frac{1}{2}}p^{4\times\frac{1}{2}}q^{-2\times\frac{1}{2}} = 3p^2q^{-1} \n3p^2q^{-1} \times 2p^{-3}q^5 = 6p^{2+(-3)}q^{-1+5} = 6p^{-1}q^4 = \frac{6q^4}{p}$.

(c) $v = \sqrt{\frac{k-3t}{t}}$. Square both sides: $v^2 = \frac{k-3t}{t} \nv^2 t = k - 3t \nv^2 t + 3t = k \nt(v^2 + 3) = k \nt = \frac{k}{v^2+3}$.

(d) $\frac{4}{x-1} + \frac{3}{x+2} = 2$. Multiply throughout by $(x-1)(x+2)$: $4(x+2) + 3(x-1) = 2(x-1)(x+2) \n4x + 8 + 3x - 3 = 2(x^2 + x - 2) \n7x + 5 = 2x^2 + 2x - 4 \n2x^2 - 5x - 9 = 0$. Using the quadratic formula with $a = 2$, $b = -5$, $c = -9$: $x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(-9)}}{2(2)} = \frac{5 \pm \sqrt{25 + 72}}{4} = \frac{5 \pm \sqrt{97}}{4} \n\sqrt{97} = 9.84886\ldots \nx = \frac{5 + 9.84886}{4} = 3.71221\ldots$ or $x = \frac{5 - 9.84886}{4} = -1.21221\ldots \nx = 3.71$ or $x = -1.21$ (correct to 2 decimal places). Check: $x = 3.71221$: $\frac{4}{2.71221} + \frac{3}{5.71221} = 1.47481 + 0.52519 = 2.0000$; $x = -1.21221$: $\frac{4}{-2.21221} + \frac{3}{0.78779} = -1.80815 + 3.80815 = 2.0000$.

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2. Mei runs a small café.

(a) Mei buys a coffee machine from Italy for €2 450. The exchange rate is €1 = S $1.48. Goods and Services Tax (GST) of 9% Calculate the total amount, in Singapore dollars, that Mei pays for the machine. [2]

(b) Mei increased the selling price of a cup of latte by 8% to $5.94.

Calculate the selling price of a cup of latte before the increase. [2]

(c) The cost of making a cup of latte is made up of the cost of coffee beans, the cost of milk and other costs in the ratio 5 : 4 : 3. The coffee beans for one cup cost $1.25.

Calculate the percentage profit Mei makes on a cup of latte sold at $5.94. [2]

(d) Mei spends $180 on Grade A coffee beans, which cost $x per kilogram.

Grade B coffee beans cost $5 per kilogram more than Grade A coffee beans.

For $180, Mei would get 3 kg less of Grade B coffee beans than of Grade A coffee beans. [5]

(i) Form an equation in x and show that it simplifies to $x^2 + 5x - 300 = 0$. [3]

(ii) Solve the equation $x^2 + 5x - 300 = 0$. [1]

(iii) Hence find the cost of 1 kilogram of Grade B coffee beans. [1]

(a) Price in Singapore dollars $= 2450 \times 1.48 = \$3626$.
With GST, total paid $= 3626 \times 1.09 = \$3952.34$.

(b) The new price is 108% of the old price.
$108\% \to \$5.94$, so $100\% \to \frac{5.94}{1.08} = \$5.50$.

(c) Beans are 5 parts of the ratio 5 : 4 : 3, which has 12 parts in all.
5 parts $= \$1.25$, so 12 parts $= \frac{1.25}{5} \times 12 = \$3.00$ (the cost of making a cup).
Profit per cup $= 5.94 - 3.00 = \$2.94$.
Percentage profit $= \frac{2.94}{3.00} \times 100\% = 98\%$.

(d)(i) Mass of Grade A beans bought for $\$180 = \frac{180}{x}$ kg.
Grade B beans cost $\$(x + 5)$ per kilogram, so the mass of Grade B beans bought for $\$180 = \frac{180}{x+5}$ kg.
The Grade B mass is 3 kg less: $\frac{180}{x} - \frac{180}{x+5} = 3$.
Multiply through by $x(x + 5)$: $180(x + 5) - 180x = 3x(x + 5)$
$180x + 900 - 180x = 3x^2 + 15x$
$900 = 3x^2 + 15x$
Divide by 3: $300 = x^2 + 5x$, so $x^2 + 5x - 300 = 0$ (shown).

(d)(ii) $x^2 + 5x - 300 = 0$
$(x - 15)(x + 20) = 0$
$x = 15$ or $x = -20$.

(d)(iii) x is a price per kilogram, so $x = -20$ is rejected and $x = 15$.
Cost of 1 kg of Grade B beans $= 15 + 5 = \$20$.
(Check: \$180 buys $\frac{180}{15} = 12$ kg of Grade A and $\frac{180}{20} = 9$ kg of Grade B; $12 - 9 = 3$ kg.)

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3. The diagram shows a sketch of the curve $y = ax^2 + bx + c$.
The curve crosses the x-axis at the points $(-1, 0)$ and $(3, 0)$ and crosses the y-axis at the point $(0, 3)$.
(a) Find the value of a, of b and of c. [3]
(b) [3]
(i) Write the equation of the curve in the form $y = k - (x - h)^2$, where h and k are integers. [2]
(ii) Hence write down the coordinates of the maximum point of the curve. [1]
(c) The point P lies on the curve and has x-coordinate 4.
Find the equation of the straight line that passes through P and the maximum point of the curve. [3]

(a) Since the curve crosses the x-axis at $x = -1$ and $x = 3$, its equation can be written as $y = a(x + 1)(x - 3)$.
The curve passes through $(0, 3)$: $3 = a(0 + 1)(0 - 3) = -3a$, so $a = -1$.
Expanding, $y = -(x + 1)(x - 3) = -(x^2 - 2x - 3) = -x^2 + 2x + 3$.
Hence $a = -1$, $b = 2$, $c = 3$.

(b)(i) $-x^2 + 2x + 3 = -(x^2 - 2x) + 3 = -\left[(x - 1)^2 - 1\right] + 3 = -(x - 1)^2 + 1 + 3 = 4 - (x - 1)^2$.
So $k = 4$ and $h = 1$.

(b)(ii) Since $(x - 1)^2 \geq 0$, the greatest value of y is 4, occurring when $x = 1$.
The maximum point is $(1, 4)$.

(c) When $x = 4$: $y = -(4)^2 + 2(4) + 3 = -16 + 8 + 3 = -5$, so P is $(4, -5)$.
Gradient of the line through $P(4, -5)$ and the maximum point $(1, 4)$: $\frac{4-(-5)}{1-4} = \frac{9}{-3} = -3$.
Using the point $(1, 4)$: $y - 4 = -3(x - 1)$, so $y = -3x + 7$.
Check at P: $-3(4) + 7 = -5$. Correct.
The equation of the line is $y = -3x + 7$.

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4. The table shows some values of x and the corresponding values of y for $y = x^2 + \frac{3}{x}$.

x	-3	-2.5	-2	-1.5	-1	-0.5	0.5	1	1.5	2	2.5	3
y	8	5.05	2.5		-2	-5.75	6.25	4	4.25	5.5		10

(a) Complete the table of values. [2]
(b) On the grid, draw the graph of $y = x^2 + \frac{3}{x}$ for $-3 \leq x \leq 3$. [3]
(c) [6]
(i) By drawing a tangent, use your graph to find the gradient of the curve at the point where $x = 2$. [2]
(ii) Use your graph to explain why the equation $x^2 + \frac{3}{x} = 3$ has only one solution. [1]
(iii) By drawing a suitable straight line on the grid, solve the equation $x^3 - 2x^2 - 3x + 3 = 0$. [3]

(a) $y = x^2 + \frac{3}{x}$\nWhen $x = -1.5$: $y = (-1.5)^2 + \frac{3}{-1.5} = 2.25 - 2 = 0.25$\nWhen $x = 2.5$: $y = 2.5^2 + \frac{3}{2.5} = 6.25 + 1.2 = 7.45$\nFull table: $x = -3, -2.5, -2, -1.5, -1, -0.5, 0.5, 1, 1.5, 2, 2.5, 3$ gives $y = 8, 5.05, 2.5, 0.25, -2, -5.75, 6.25, 4, 4.25, 5.5, 7.45, 10$.\n\n(b) Plot the twelve points on the grid (2 cm to 1 unit on the x-axis, 1 cm to 1 unit on the y-axis, $-3 \leq x \leq 3$, $-6 \leq y \leq 10$) and join them with two smooth curves: the left branch from $(-3, 8)$ down to $(-0.5, -5.75)$, and the right branch from $(0.5, 6.25)$ down to a minimum near $(1.1, 3.9)$ and up to $(3, 10)$. No curve is drawn between $x = -0.5$ and $x = 0.5$ because y is undefined at $x = 0$.\n\n(c)(i) Draw the tangent to the curve at $(2, 5.5)$ and take two points on it far apart, e.g. $(1, 2.25)$ and $(3, 8.75)$.\nGradient $= \frac{8.75 - 2.25}{3 - 1} = \frac{6.5}{2} = 3.25$\n(Exact value: $\frac{dy}{dx} = 2x - \frac{3}{x^2} = 4 - 0.75 = 3.25$; accept 2.9 to 3.6 from a drawn tangent.)\n\n(c)(ii) The solutions of $x^2 + \frac{3}{x} = 3$ are the x-coordinates of the points where the curve meets the line $y = 3$. The line $y = 3$ cuts the left-hand branch once, at $x \approx -2.1$, but the right-hand branch has its lowest point at $y \approx 3.9$ (exactly $y = 3.93$ at $x = 1.14$), which is above $y = 3$, so the line does not meet the right-hand branch. Hence there is only one solution.\n\n(c)(iii) $x^3 - 2x^2 - 3x + 3 = 0$\nDivide throughout by x ($x \neq 0$): $x^2 - 2x - 3 + \frac{3}{x} = 0$\n$x^2 + \frac{3}{x} = 2x + 3$\nSo the required line is $y = 2x + 3$, which passes through $(-3, -3)$, $(0, 3)$ and $(3, 9)$.\nDraw the line on the grid and read the x-coordinates of the three points where it cuts the curve:\n$x = -1.45$, $x = 0.75$, $x = 2.7$\n(Exact values $-1.4605\ldots$, $0.7609\ldots$, $2.6996\ldots$; check: at $x = 2.7$, $x^2 + \frac{3}{x} = 7.29 + 1.111 = 8.40$ and $2x + 3 = 8.40$. Accept $-1.55 \leq x \leq -1.4$, $0.7 \leq x \leq 0.85$, $2.6 \leq x \leq 2.8$.)

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5. The diagram shows a sector OAB of a circle, centre O and radius 9 cm.
Angle $AOB = 1.2$ radians.
C is the point on OB such that $OC = 5$ cm.
The region bounded by AC, CB and the arc AB is shaded.
(a) Calculate AC. [2]
(b) Calculate angle OAC.
Give your answer in degrees. [2]
(c) Calculate [?]
(i) the perimeter of the shaded region, [2]
(ii) the area of the shaded region. [3]

The calculator is set to radians for every step that uses the sector angle.

(a) In triangle OAC, $OA = 9$ cm, $OC = 5$ cm and the included angle $AOC = 1.2$ radians.
By the cosine rule,
$AC^2 = 9^2 + 5^2 - 2(9)(5)\cos 1.2$
$= 106 - 90(0.362357\ldots)$
$= 106 - 32.6121\ldots$
$= 73.3878\ldots$
$AC = \sqrt{73.3878\ldots} = 8.56666\ldots = 8.57$ cm (3 s.f.)

(b) In triangle OAC, by the sine rule,
$\frac{\sin OAC}{OC} = \frac{\sin AOC}{AC}$
$\sin OAC = \frac{5 \sin 1.2}{8.56666\ldots} = \frac{5(0.932039\ldots)}{8.56666\ldots} = 0.543991\ldots$
Angle OAC is opposite the shortest side OC, so it is acute.
Angle $OAC = \sin^{-1}(0.543991\ldots) = 32.9557\ldots^\circ = 33.0^\circ$ (1 d.p.)

(c)(i) Arc $AB = r\theta = 9 \times 1.2 = 10.8$ cm
$CB = OB - OC = 9 - 5 = 4$ cm
Perimeter of shaded region = arc $AB + CB + AC$
$= 10.8 + 4 + 8.56666\ldots$
$= 23.3666\ldots = 23.4$ cm (3 s.f.)

(c)(ii) Area of sector $OAB = \frac{1}{2}r^2\theta = \frac{1}{2}(9^2)(1.2) = 48.6$ cm^2
Area of triangle $OAC = \frac{1}{2}(OA)(OC)\sin AOC = \frac{1}{2}(9)(5)\sin 1.2 = 20.9708\ldots$ cm^2
Shaded area $= 48.6 - 20.9708\ldots = 27.6291\ldots = 27.6$ cm^2 (3 s.f.)

6. The points A, B and C have coordinates $(-4,\ -1)$, $(2,\ 7)$ and $(10,\ -1)$ respectively.

(a) Find the equation of the line which passes through C and is parallel to AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers. [3]

(b) Calculate the length of AB. [2]

(c) Calculate the area of triangle ABC. [2]

(d) OPQ is a triangle.
$\overrightarrow{OP} = \mathbf{p}$ and $\overrightarrow{OQ} = \mathbf{q}$.
R is the point on PQ such that $PR : RQ = 1 : 2$.
S is the point on OQ such that $OS : SQ = 1 : 2$. [4]

(i) Express $\overrightarrow{OR}$ in terms of $\mathbf{p}$ and $\mathbf{q}$, as simply as possible. [2]

(ii) Express $\overrightarrow{SR}$ in terms of $\mathbf{p}$ and $\mathbf{q}$, as simply as possible. [1]

(iii) Hence find $\dfrac{\text{area of triangle } QSR}{\text{area of triangle } QOP}$. [1]

(a) Gradient of $AB = \frac{7-(-1)}{2-(-4)} = \frac{8}{6} = \frac{4}{3}$
The required line is parallel to AB, so its gradient is also $\frac{4}{3}$, and it passes through $C(10,\ -1)$:
$y - (-1) = \frac{4}{3}(x - 10)$
$3(y + 1) = 4(x - 10)$
$3y + 3 = 4x - 40$
$4x - 3y = 43$
Check with C: $4(10) - 3(-1) = 40 + 3 = 43$.

(b) $AB = \sqrt{(2-(-4))^2 + (7-(-1))^2} = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ units

(c) $A(-4,\ -1)$ and $C(10,\ -1)$ have the same y-coordinate, so AC is horizontal.
Base $AC = 10 - (-4) = 14$ units.
The perpendicular height from B to AC is $7 - (-1) = 8$ units.
Area of triangle $ABC = \frac{1}{2} \times 14 \times 8 = 56$ units^2

(d)(i) $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \mathbf{q} - \mathbf{p}$
Since $PR : RQ = 1 : 2$, $\overrightarrow{PR} = \frac{1}{3}\overrightarrow{PQ} = \frac{1}{3}(\mathbf{q} - \mathbf{p})$
$\overrightarrow{OR} = \overrightarrow{OP} + \overrightarrow{PR} = \mathbf{p} + \frac{1}{3}\mathbf{q} - \frac{1}{3}\mathbf{p} = \frac{2}{3}\mathbf{p} + \frac{1}{3}\mathbf{q}$

(d)(ii) Since $OS : SQ = 1 : 2$, $\overrightarrow{OS} = \frac{1}{3}\mathbf{q}$
$\overrightarrow{SR} = \overrightarrow{OR} - \overrightarrow{OS} = \frac{2}{3}\mathbf{p} + \frac{1}{3}\mathbf{q} - \frac{1}{3}\mathbf{q} = \frac{2}{3}\mathbf{p}$

(d)(iii) $\overrightarrow{SR} = \frac{2}{3}\overrightarrow{OP}$, so SR is parallel to OP and $SR = \frac{2}{3}OP$.
Triangles QSR and QOP share the angle at Q, and angle QSR = angle QOP (corresponding angles, $SR \parallel OP$), so the triangles are similar with $\frac{SR}{OP} = \frac{2}{3}$.
$\frac{\text{area of triangle } QSR}{\text{area of triangle } QOP} = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$
Check by common heights: triangles QSR and QOR have the common vertex R with bases QS and QO on the same line, so $\frac{\text{area } QSR}{\text{area } QOR} = \frac{QS}{QO} = \frac{2}{3}$; triangles QOR and QOP have the common vertex O with bases QR and QP on the same line, so $\frac{\text{area } QOR}{\text{area } QOP} = \frac{QR}{QP} = \frac{2}{3}$. Hence $\frac{\text{area } QSR}{\text{area } QOP} = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$, as before.

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7. The table shows the distribution of the distances, d km, cycled by the members of a cycling club in one week.

Distance (d km)	$0 < d \leqslant 20$	$20 < d \leqslant 40$	$40 < d \leqslant 60$	$60 < d \leqslant 80$	$80 < d \leqslant 100$
Frequency	5	x	18	13	7

The estimated mean distance cycled is 50 km.

(a) Find the value of x. [2]

(b) Calculate an estimate of the standard deviation of the distances. [2]

(c) [4]

(i) One of the members is chosen at random. Find the probability that this member cycled more than 60 km. [1]

(ii) Two of the members are chosen at random. Find the probability that exactly one of them cycled more than 60 km. Give your answer as a fraction in its lowest terms. [3]

(a) Mid-values 10, 30, 50, 70, 90.
$\sum f = 5 + x + 18 + 13 + 7 = 43 + x$
$\sum fx = 5(10) + 30x + 18(50) + 13(70) + 7(90) = 50 + 30x + 900 + 910 + 630 = 2490 + 30x$
Estimated mean $= \frac{2490 + 30x}{43 + x} = 50$
$2490 + 30x = 2150 + 50x$
$20x = 340$
$x = 17$

(b) Total frequency $= 43 + 17 = 60$.
$\sum fx^2 = 5(100) + 17(900) + 18(2500) + 13(4900) + 7(8100) = 500 + 15300 + 45000 + 63700 + 56700 = 181200$
Standard deviation $= \sqrt{\frac{181200}{60} - 50^2} = \sqrt{3020 - 2500} = \sqrt{520} = 22.803\ldots = 22.8$ km (3 s.f.).

(c)(i) Number who cycled more than 60 km $= 13 + 7 = 20$ out of 60.
$P(\text{more than 60 km}) = \frac{20}{60} = \frac{1}{3}$.

(c)(ii) Selection is without replacement. 20 members cycled more than 60 km and 40 did not.
$P(\text{exactly one}) = P(\text{more, not more}) + P(\text{not more, more}) = \frac{20}{60} \times \frac{40}{59} + \frac{40}{60} \times \frac{20}{59} = 2 \times \frac{800}{3540} = \frac{1600}{3540} = \frac{80}{177}$.

8. A, B, C and D are points on a circle, centre O.
AOC is a diameter of the circle.
The tangent to the circle at C meets AB produced at T.
Angle $BAC = 32°$ and angle $CBD = 37°$.
(a) Find, giving a reason for each step of your working, [5]
(i) angle ACB, [2]
(ii) angle BCT, [2]
(iii) angle CAD. [1]
(b) Explain why AD is not parallel to BC. [1]
(c) The radius of the circle is 6 cm. [4]
(i) Show that triangle ABC is similar to triangle CBT.
Give a reason for each statement you make. [2]
(ii) Calculate BT. [2]

(a)(i) Angle $ABC = 90°$ (angle in a semicircle, since AC is a diameter).
Angle $ACB = 180° - 90° - 32° = 58°$ (angle sum of triangle ABC).

(a)(ii) Angle $OCT = 90°$ (tangent perpendicular to radius OC at the point of contact C), so angle $ACT = 90°$ since AOC is a straight line.
Angle $BCT = 90° - 58° = 32°$.

(a)(iii) Angle CAD = angle $CBD = 37°$ (angles in the same segment, on chord CD).

(b) If AD were parallel to BC, then angle CAD would equal angle ACB (alternate angles). But angle $CAD = 37°$ and angle $ACB = 58°$, and $37° \neq 58°$, so AD is not parallel to BC.

(c)(i) In triangles ABC and CBT:
angle $ABC = 90°$ (angle in a semicircle) and angle $CBT = 180° - 90° = 90°$ (angles on the straight line ABT), so angle ABC = angle CBT;
angle $BAC = 32°$ (given) and angle $BCT = 32°$ (from (a)(ii)), so angle BAC = angle BCT.
Two pairs of corresponding angles are equal, so triangle ABC is similar to triangle CBT (AA).

(c)(ii) $AC = 2 \times 6 = 12$ cm.
In right-angled triangle ABC: $BC = 12 \sin 32° = 6.359\ldots$ cm and $AB = 12 \cos 32° = 10.176\ldots$ cm.
From the similar triangles, $\frac{BT}{BC} = \frac{CB}{AB}$, so $BT = \frac{BC^2}{AB} = \frac{(12 \sin 32°)^2}{12 \cos 32°} = 3.973\ldots$
(Equivalently, in right-angled triangle CBT, $BT = BC \tan 32° = 6.359\ldots \times \tan 32° = 3.973\ldots$.)
$BT = 3.97$ cm (3 s.f.).

Consistency check: with O at the origin and radius 6, $A(-6, 0)$, $C(6, 0)$, B at $64°$ from OC (angle at centre $= 2 \times 32°$), D at $-74°$ (angle at centre $= 2 \times 37°$); line AB has gradient $\tan 32°$ and meets the tangent $x = 6$ at $T(6, 12 \tan 32°) = (6, 7.50)$, above B, so ABT is a straight line with B between A and T. Every given angle is realised.

9. Mr Tan is deciding whether to install a rainwater tank to water the plants in his garden.
He finds the following information.

Rainwater tank
The tank is a cylinder with internal diameter 80 cm and internal height 120 cm.
Price, including installation: \$420
Expected lifespan: 10 years

Collecting rainwater
Rain falling on the roof of the garden shed drains into the tank.
The roof is a rectangle 3.6 m by 2.5 m.
80% of the rain that falls on the roof reaches the tank.
Average rainfall: 175 mm per month

Watering the garden
Mr Tan waters his garden with a hose for 12 minutes every day.
The hose delivers 9 litres of water per minute.

Monthly water tariff

First 40 m^3	\$2.74 per m^3
Each m^3 above 40 m^3	\$3.69 per m^3

1 m^3 = 1000 litres
(a) Calculate the capacity of the tank.
Give your answer in litres. [2]
(b) In a month with 30 days, calculate [3]
(i) the volume of water, in litres, Mr Tan uses to water his garden, [1]
(ii) the volume of rainwater, in litres, that reaches the tank. [2]
(c) Before installing the tank, Mr Tan's household uses 43 m^3 of tap water each month.
Assume that all the rainwater that reaches the tank is used to water the garden instead of tap water.
Is it worth Mr Tan installing the rainwater tank?
Justify your answer with calculations. [5]

(a) Radius = 40 cm. Capacity $= \pi \times 40^2 \times 120 = 192\,000\pi = 603\,185.8$ cm^3. Since 1000 cm^3 = 1 litre, capacity = 603.19 = 603 litres (3 s.f.). [M1 for $\pi r^2 h$ with $r = 40$, A1]

(b)(i) Water used per day $= 9 \times 12 = 108$ litres. In 30 days: $108 \times 30 = 3240$ litres. [B1]

(b)(ii) Roof area $= 3.6 \times 2.5 = 9$ m^2. Rainfall 175 mm = 0.175 m, so volume of rain on the roof in a month $= 9 \times 0.175 = 1.575$ m^3 = 1575 litres. Volume reaching the tank $= 80\% \times 1575 = 1260$ litres. [M1 for 9×0.175 (or 1575), A1]

(c) Rainwater available = 1260 litres per month, which is less than the 3240 litres used, so all 1.26 m^3 replaces tap water each month. [M1 for 1.26 m^3 replacing tap water]
Tap water used falls from 43 m^3 to $43 - 1.26 = 41.74$ m^3, still above 40 m^3, so the water saved is all charged at the higher rate of \$3.69 per m^3. [M1 for recognising the saving is at the \$3.69 tier]
Monthly saving $= 1.26 \times 3.69 = \$4.6494 = \$4.65$. [A1]
Saving over the 10-year lifespan $= 4.6494 \times 12 \times 10 = \$557.93$ (or $4.65 \times 120 = \$558.00$). [M1]
$\$557.93 > \$420$, so the tank saves more than it costs: net saving $\approx$ \$138 over 10 years, and it pays for itself after $420 \div 4.6494 = 90.3$, i.e. 91 months (about 7.5 years), within its lifespan. Conclusion: YES, it is worth installing. [A1 for a conclusion consistent with correct figures]
Alternative acceptable conclusion: NO — the net saving of about \$138 over 10 years (about \$14 a year) is small, and rainfall varies so the tank may overflow in wet months and run dry in dry months; award the final A1 for this if the figures are correct and the reason is valid.
Note: using \$2.74 per m^3 gives $1.26 \times 2.74 = \$3.45$ per month and \$414.29 over 10 years, which is less than \$420; this uses the wrong tier, so award M1 M0 A0 M1 A0 (max 2).